macpan2 calibration

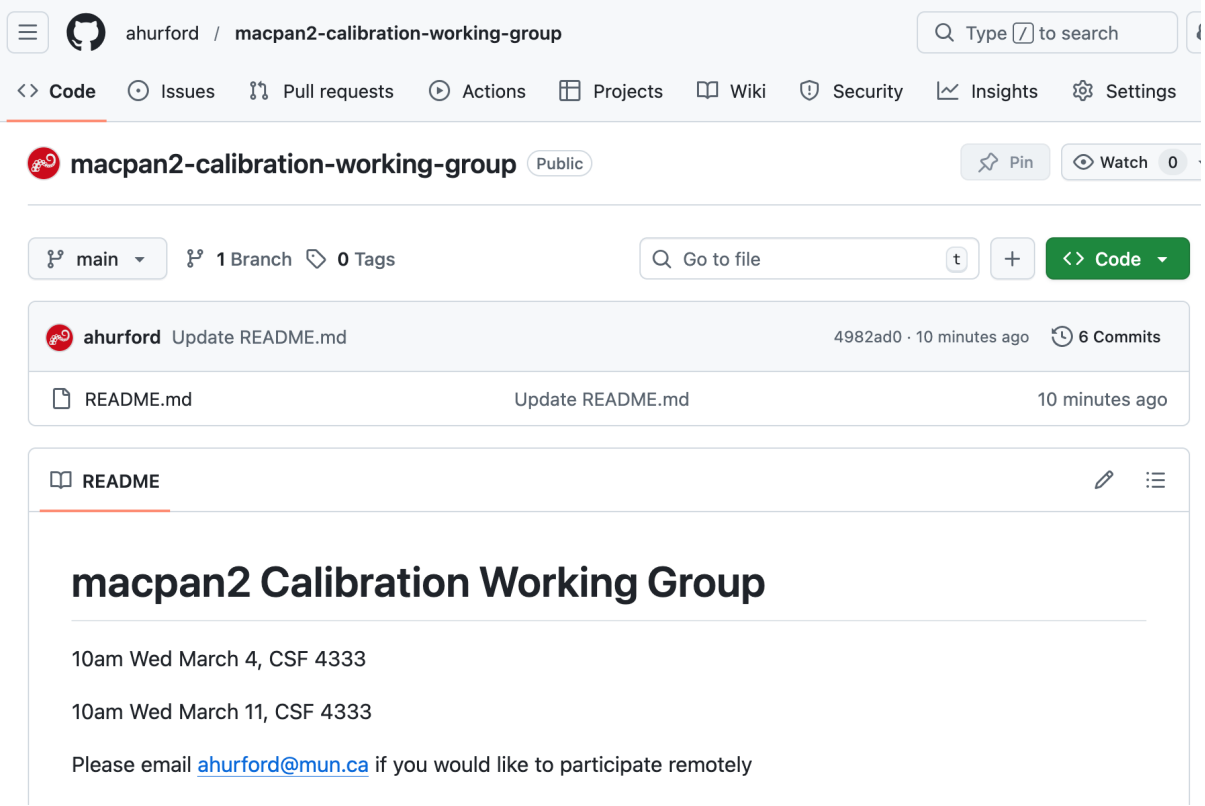
Amy Hurford

March 4, 2026

MEMORIAL UNIVERSITY

ACCDMi
Atlantic Canada Centre for Disease Ecology Modelling

Working group repository



Link

MEMORIAL UNIVERSITY

ACCDMi

macpan2

`macpan2` is a software platform for building and calibrating compartmental models of infectious disease. It supports flexible model specification and fast parameter calibration, making it easier for modellers to respond to emerging public health threats. Developed through collaboration with the Public Health Agency of Canada (PHAC), `macpan2` is being used to support responses to diseases such as mpox (Milwid et al. 2023), measles (PHAC 2024, interactive measles model), and COVID-19 (Simmons et al. 2025; Miranda et al. 2025), and internal work on pandemic preparedness. It has been used for a retrospective analysis of COVID-19 in Newfoundland (Anokye et al. 2025).

Link Link to “why macpan2?” video

MEMORIAL UNIVERSITY

ACCDMi
Atlantic Canada Centre for Disease Ecology Modelling

Goal: calibration of mechanistic models

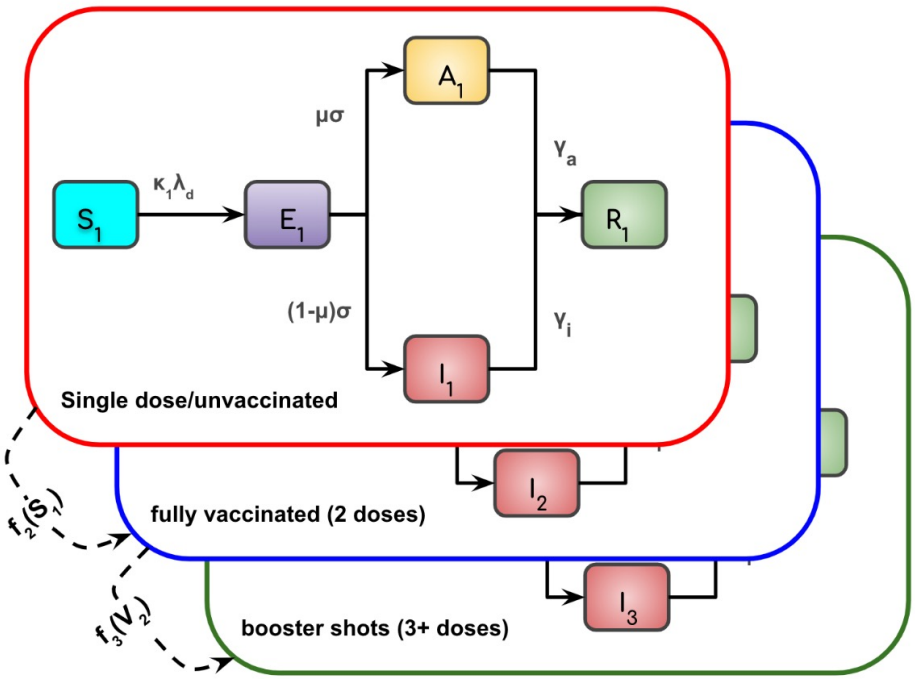


Fig 2. The SEAIR model with three vaccination cohorts. The vaccination cohorts are: unvaccinated/single-dose (cohort 1; red), fully vaccinated (cohort 2; blue), and booster-vaccinated (cohort 3; green). Each cohort contains susceptible, exposed, infectious (symptomatic and asymptomatic), and recovered compartments. Solid arrows show disease progression; dashed arrows show vaccination transitions $f_2(S_1)$ (cohort 1 to cohort 2) and $f_3(V_2)$ (cohort 2 to cohort 3).

- Mechanistic models describe processes
- Contrast with pattern matching approaches

Examples are from Anokye et al. MedRxiv. link

MEMORIAL UNIVERSITY

ACCDMi

Parameter estimation for mechanistic models

Symbol	Definition	Value	Details
ζ	Relative infectiousness of asymptomatic cases ($0 \leq \zeta \leq 1$)	$\zeta = 0.75$	(29)
κ_1	Susceptibility, unvaccinated	$\kappa_1 = 1.0$	reference value
$v_2(t), v_3(t)$	Daily vaccination rates (2-dose & booster)	vaccination data	Sec. S1.7 in S1 Text
Fitted parameters			
σ	Rate from exposed to infectious	0.356 days^{-1} (CrI: [0.243, 0.469])	[1/4,1/2] days^{-1} (30)
γ_a	Recovery rate, asymptomatic	0.106 days^{-1} (CrI: [0.090, 0.123])	[1/11,1/8] days^{-1} (31)
γ_i	Recovery rate, symptomatic	0.144 days^{-1} (CrI: [0.124, 0.164])	[1/8,1/6] days^{-1} (31)
κ_2	Relative susceptibility, 2-dose vaccinated ($0 \leq \kappa_2 \leq 1$)	$\kappa_2 = 0.911$ (CrI: [0.862, 0.960])	[0.85, 0.96] (32)
κ_3	Relative susceptibility, booster vaccinated ($0 \leq \kappa_3 \leq 1$)	$\kappa_3 = 0.303$ (CrI: [0.229, 0.376])	[0.20, 0.40] (32)
$\beta(t)$	Time-varying transmission rate (Eqn 2)	$\beta(t)$ is a component of $\mathcal{R}_c$ (see Eqn 3), which is shown in Figure 5.	Discussed in detail in the Methods subsection 'Time-varying transmission rate'.

Fixed parameters: a point estimate based on published values

Fitted parameters: assumed prior distributions based on published values. Estimated using MAP

MEMORIAL UNIVERSITY

ACCDMi
Atlantic Canada Centre for Disease Ecology Modelling

Maximum a posteriori (MAP) estimation

MAP Estimation

The recommended approach to model fitting with `macpan2` is maximum a posteriori (MAP) estimation, which finds the values of parameters at the peak of a posterior distribution. This approach allows us to get approximate Bayesian uncertainty estimates by approximating the posterior distribution as a multivariate Gaussian centered at the peak obtained by MAP. One is able to declare that parameters be considered random effects, which means that they are integrated out of the posterior distribution using the Laplace approximation. Using this approach allows us to reduce the dimensionality of the posterior distribution used in MAP by only optimizing over the fixed effect parameters.

link

link about maximum a posteriori (MAP) estimation

MEMORIAL UNIVERSITY

ACCDMi

Prior and posterior distributions

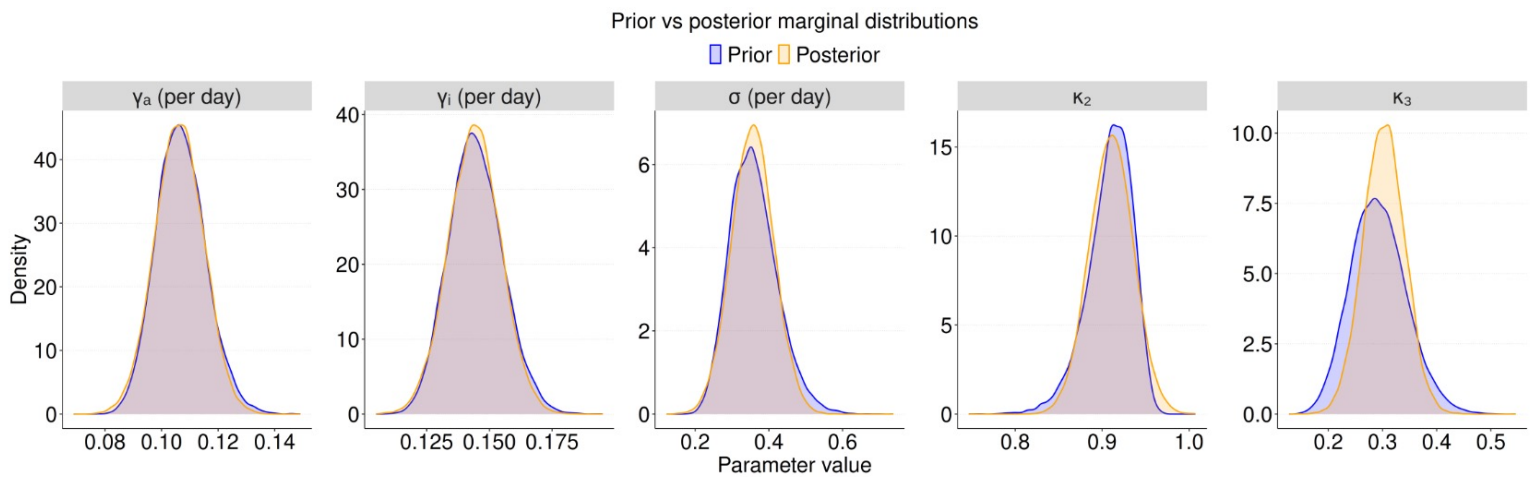


Fig. S3: **Prior and posterior marginal distributions of fitted model parameters.** Blue curves show prior distributions and orange curves show posterior distributions. Panels correspond to the recovery rate of asymptomatic infections (γ_a, per day), the recovery rate of symptomatic infections (γ_i, per day), the rate of progression from exposed to infectious (σ, per day), the relative susceptibility of double-vaccinated individuals (κ_2), and the relative susceptibility of boosted individuals (κ_3).

Statistical model

Statistical Model

Above we were not specific about the statistical model used to fit the data. Here we describe it.

Let the observed and simulated trajectories be vectors I_{obs} and I_{sim}. The I symbol is chosen because we fitted to prevalence above, but it could be any trajectory in the model. For example, `traj = "infection"` would have fitted to incidence, because the `infection` variable in the model is the number of new cases at every time step.

The simulated trajectories are actually a function of the vector, $\mathbf{b}$, of default values that we chose to make statistical parameters. Therefore, we write the simulated trajectory as a function, $I_{\text{sim}}(\mathbf{b})$.

We assume that the observed trajectory is Poisson distributed with mean given by the simulated trajectory.

$$I_{\text{obs}} \sim \text{Poisson}(I_{\text{sim}}(\mathbf{b}))$$

Given these assumptions we choose $\mathbf{b}$ to maximize the resulting likelihood function, and use functionality from the `TMB` package (and sometimes the `tmbstan` / `rstan` packages) to do statistical inference on the fitted parameters and trajectories.

This statistical model will often be too simple. The `macpan2` package has an extremely flexible developer interface that allows for more detailed control over `TMB`, `tmbstan`, and `rstan`. This interface allows for arbitrary likelihood functions, prior distributions, parameter transformations, flexible parameter time-variation models, random effects and more. See here and here for more information, although because these guides describe a developer interface the instructions may be unclear to some or many readers. Our plan is to continue adding interface layers, such as the interface described in this vignette, so that more of `macpan2` can be exposed to users.

link

MEMORIAL UNIVERSITY

ACCDMi
Atlantic Canada Centre for Disease Ecology Modelling

Statistical model

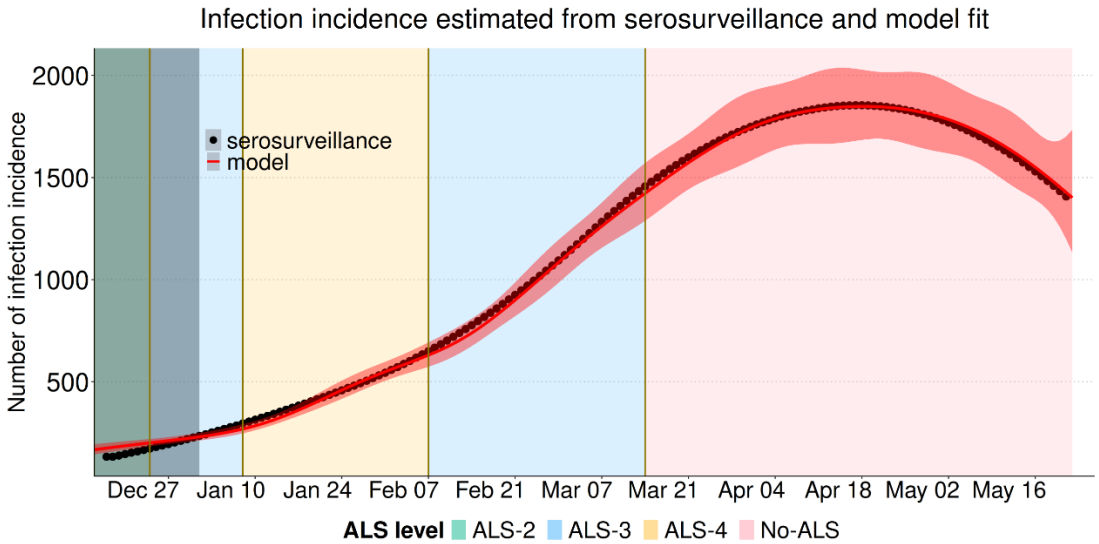


Fig 3. Model fit to infection incidence as calculated from serological prevalence data for December 15, 2021–May 22, 2022. Model-estimated incidence (red line, shaded band is the 95% confidence interval) fit to the seroincidence data (black dots). Background shaded colors indicate different levels of NPIs under the ALS system (from least to most strict): No-ALS (pink); ALS-2 (pale green), ALS-3 (pale blue); and ALS-4 (pale yellow). The solid vertical line indicates the date on which the public health emergency declaration that instituted the ALS was repealed.

macpan2 uses template model builder (TMB) to calibrate models to data

Link about what latent variable models are

MEMORIAL UNIVERSITY

ACCDMi

See also topics from workshop materials

macpan-workshop

Workshop on `macpan2` – Companion Materials

Author: Steve Walker

Last Updated: 2025-03-20

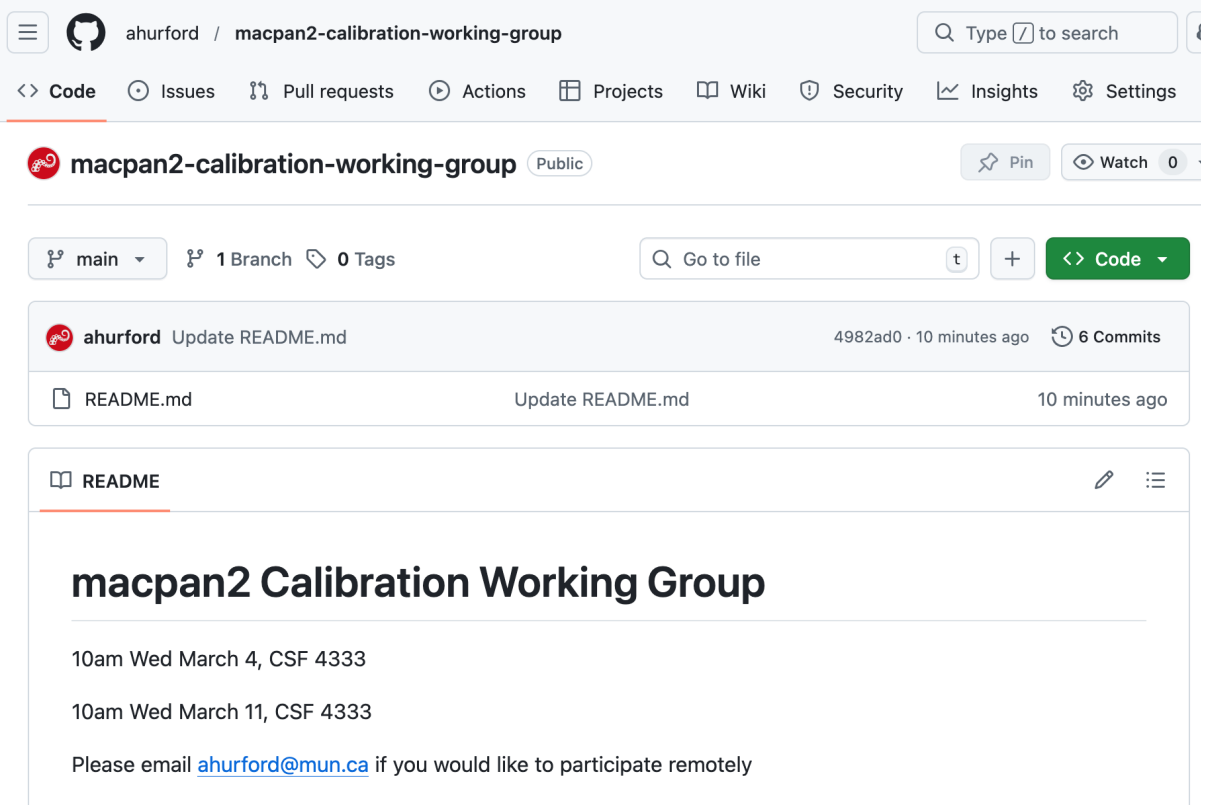
Link to macpan2 workshop

- Exploration
  - Starter Model Library
  - Simulating Dynamics
  - Relating Model Specifications to Box Diagrams
  - Relating Model Specifications to Explicit Dynamics
  - Basics of Modifying Model Specifications
  - Compare Simulated and Observed Incidence
  - Reporting Delays
  - Hospital Admissions and Occupancy
  - Symptom Status
  - Seroprevalence
  - Vaccination
  - Wastewater Pathogen Concentrations
  - Vital Dynamics
  - Importation and Stochasticity

MEMORIAL UNIVERSITY

ACCDMi

To get started visit the working group repository



Link

MEMORIAL UNIVERSITY

ACCDMi